Relnl algebra - theoretical query language
- "assembly lang of DBs"

<u>Operations</u>:

(selection)

"Select" - returns a subset of tuples/rows in a relation

$$\sigma_{condition}(R)$$

which rows you are keeping

$$\sigma_{age < 18}(Students)$$

↳ a new reln w/ fewer rows

"Project" (projection)
- returns a subset of columns/attrs in a relation

students

sid	age	name

$\Pi_{sid, age}$ (Students)

↳

sid	age

In pure relnl model, relns are <u>sets</u>

→ no dups

<u>set operations</u>

Union (∪)

Intersection (∩)

Set Difference (−)

A ∪ B → all tuples from both relations

... but again, dups are removed

These 3 ops are only valid on 2 relns w/ same schema

Cross-product / Cartesian product

$R \times S$

A new reln, w/ cols from both, where all tuples are paired in all combinations

R

sid	name
1	Dave
2	Evan
3	Rachel

S (cars)

cid	color
10	red
11	green
12	blue

$R \times S \rightarrow$

sid	name	cid	color
1	Dave	10	red
1	Dave	11	green
1	Dave	12	blue
2	Evan	10	red

...

<u>Join</u>: cross-product followed by a selection.

$$R \bowtie_{cond} S = \sigma_{cond}(R \times S)$$

R

sid	name
1	Dave
2	Evan
3	Rachel

S

sid	salary
1	\$1 MM
3	-\$3
1	\$12

$$R \bowtie_{R.sid = S.sid} S$$

R.sid	S.sid	name	salary
1	1	Dave	\$1MM
1	1	Dave	\$12
3	3	Rachel	-\$3

If no condition is specified:
"equijoin" - test equality on all attrs w/ same name.

Recipes w/ ingred iid 1?

$$\pi_{R.name}\left(\sigma_{A.iid=1}\left(R \bowtie A\right)\right)$$

all ingred ids for each recipe

just rows for ingred iid 1

recipe name

Names of recipes w/ at least one ingred at least 100 calories

$$\pi_{R.name}\left(\pi_{i,iid}\left(\sigma_{Cal>=100}(I) \bowtie A \bowtie R\right)\right)$$

just an iid

ingredients we want

those ingreds - matched to recipes

nothing

Or, let's try doing this right

$$\Big(\underbrace{\pi_{I.iid}\big(\sigma_{cal>=100}(I)\big)}_{\text{all ingreds of interest}}$$

ids of ingredients

$$\bowtie A \bowtie R\Big)$$

all recipes with ingreds of interest

$$\pi_{R.name}\Big(\qquad\qquad\Big)$$

gets up name of recipes

Eggs/Sugar example

$\sigma_{name = 'Eggs' \wedge name = 'Sugar'}$ (I)

Find all rows where within the row, name is both Eggs an Sugar.